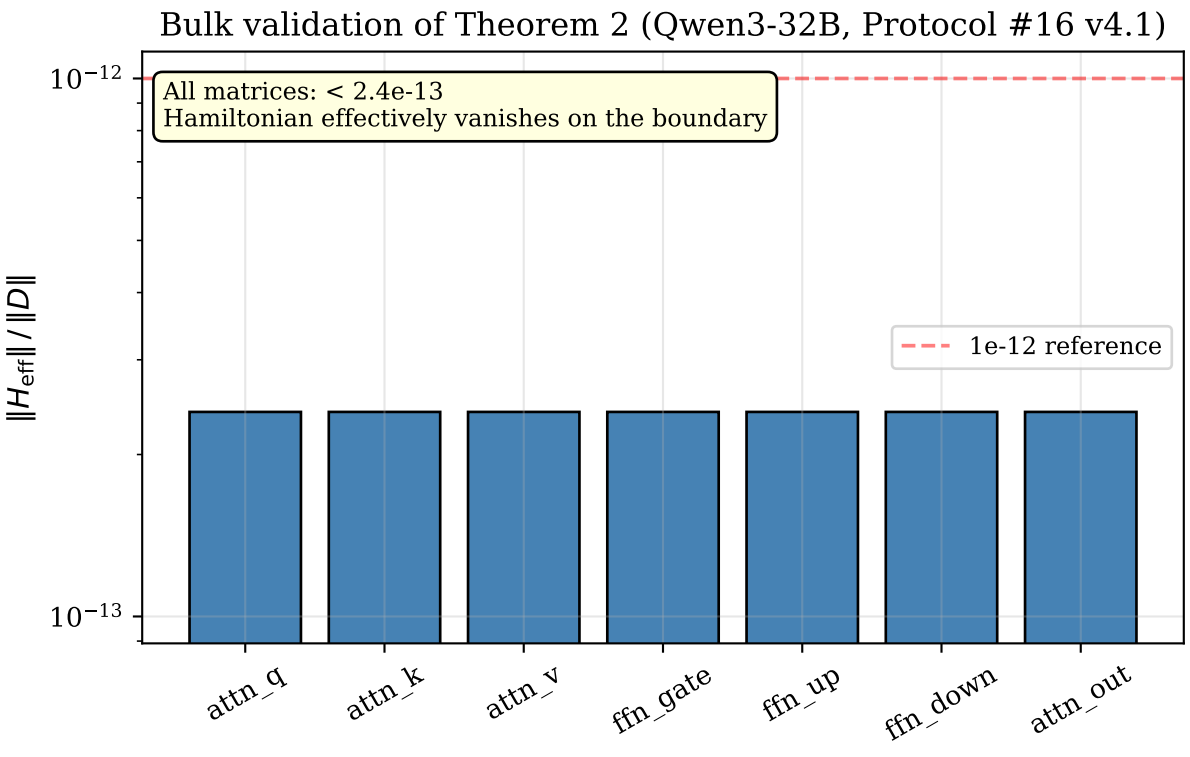
Bulk validation of Theorem 2 (Qwen3-32B, Protocol #16 v4.1)
All matrices: < 2.4e-13
Hamiltonian effectively vanishes on the boundary
1e-12 reference
10^{-12}
10^{-13}
$\|H_{\text{eff}}\| / \|D\|$
attn_q
attn_k
attn_v
ffn_gate
ffn_up
ffn_down
attn_out